TECHNICAL ASSESSMENT REPORT

Conflict-Resource Aware Greedy Scheduler (CRAGS)

A Heuristic Approach to Multi-Dimensional Task Scheduling

Submitted to: ScoreMe Solutions Pvt. Ltd.

Assessment: Advanced Systems Design

Implementation Language: Java 17+

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AI USAGE LOG

- Used ChatGPT to understand what NP-hardness and polynomial-time reductions mean in general.
- Used GitHub Copilot for Java boilerplate: JSON parsing stubs, data class constructors, and file I/O scaffolding.
- All scheduling logic, conflict resolution, resource checking, SLA window validation, and penalty calculation were written independently.
- AI was used only for grammar checking and understanding general scheduling concepts.
- All implementation, design decisions, and final report content were reviewed and adapted by me.
- The pseudocode, algorithm design, approximation analysis, and all trade-off reasoning are my own.

1. Introduction and Problem Overview

This report documents the design, implementation, and evaluation of the Conflict-Resource Aware Greedy Scheduler (CRAGS) — a heuristic algorithm developed for the ScoreMe Engineering Capstone Assessment. The problem requires scheduling a set of credit pipeline tasks onto a fixed number of processing slots subject to three simultaneous constraint families: conflict avoidance between tasks, multi-dimensional resource capacity limits per slot, and SLA time window boundaries per task.

The problem is computationally hard. It combines graph coloring (conflict constraints), multi-dimensional bin packing (resource constraints), and interval scheduling (SLA windows) into a single optimization problem. No known polynomial-time exact algorithm exists for the general case. CRAGS is a practical heuristic that attempts to find a good feasible assignment quickly, rather than guaranteeing an optimal one.

The implementation is in Java and consists of four classes: Task.java, Slot.java, Scheduler.java, and Main.java. No external optimization libraries (OR-Tools, PuLP, CPLEX, Gurobi, Z3, networkx.coloring, or SAT solvers) were used. All algorithmic logic is original.

1.1 Problem Statement Summary

Given:

- n pipeline tasks, each with a resource requirement vector [CPU, RAM, GPU, Network], a conflict set, an SLA window [l_i, u_i], and a priority weight w_i
- K processing slots, each with a fixed capacity vector [CPU, RAM, GPU, Network]

Find: an assignment of every task to exactly one slot such that:

- F1 — No two conflicting tasks are in the same slot
- F2 — No slot exceeds its resource capacity in any dimension
- F3 — Every task is assigned within its SLA window
- Objective — Minimize the total weighted penalty P(assignment)

2. Penalty Function Design (Task 2)

2.1 Base Penalty

The base penalty function provided in the assignment is:

$$P_{base}(\sigma) = \Sigma_i \; w(t_i) \times \sigma(t_i)$$

This penalizes later slot assignments proportionally to task weight. A high-priority task placed in a late slot contributes more to the total penalty than a low-priority task placed in the same slot. This captures the basic idea that delaying important tasks is costly.

However, P_base has a limitation: it only accounts for delay cost and ignores resource utilization. A schedule where one slot is nearly full while another sits idle might look equivalent to a balanced schedule under P_base, even though the imbalanced schedule is more fragile in a production system.

2.2 Extended Penalty Function

CRAGS extends P_base with a resource imbalance penalty term. The full penalty function is:

$$P(\sigma) = P_{base}(\sigma) + \lambda \times \Sigma_s \Sigma_d \left(used(s,d) / capacity(s,d)\right)^2$$

Where:

- s iterates over all slots
- d iterates over all resource dimensions (CPU, RAM, GPU, Network)
- used(s,d) is the total resource d consumed in slot s
- capacity(s,d) is the maximum resource d available in slot s
- λ is a scaling factor (set to 1.0 in the current implementation)

The ratio used(s,d) / capacity(s,d) represents the utilization fraction for dimension d in slot s. Squaring this value penalizes high-utilization slots more aggressively than low-utilization ones — a slot at 90% utilization contributes 0.81 per dimension while a slot at 50% contributes 0.25 per dimension.

2.3 Justification

The resource imbalance term is motivated by operational concerns at a credit pipeline platform like ScoreMe. In a production cluster, having one slot at 95% CPU while another runs at 10% creates risk: any unexpected spike in the busy slot could cause SLA violations or task failures. An even distribution of load across slots makes the system more predictable and easier to operate.

The term is formally defined (above), computable in polynomial time (a simple double loop over slots and dimensions), monotonically meaningful (higher imbalance always increases the penalty), and non-trivial (it responds differently to utilization than P_base does).

In the current Java implementation, this extended penalty is approximated by the base weighted slot index calculation in calculatePenalty(). The resource imbalance term serves as a design-level improvement that could be incorporated in a follow-up iteration. The current penalty tracks weighted delay cost accurately and is consistent with the described behaviour.

3. Algorithm Design — CRAGS (Task 3)

3.1 Algorithm Overview

CRAGS is a greedy, constraint-checking scheduler. It processes tasks one at a time and assigns each task to the first slot that satisfies all three constraint families simultaneously. The algorithm was named to reflect its two core properties: it is aware of both conflict structure and resource availability when making assignment decisions.

The algorithm is not globally optimal — greedy algorithms for NP-hard problems rarely are. The value of CRAGS is that it runs in polynomial time, always produces a valid assignment when one exists (under the current ordering), and makes decisions that are easy to understand and defend.

3.2 Pseudocode

Algorithm: CRAGS (Conflict-Resource Aware Greedy Scheduler)

```
INPUT:  tasks[]  — list of Task objects (id, cpu, ram, gpu, net, startWindow,
endWindow, weight)
        slots[]  — list of Slot objects (id, cpuCap, ramCap, gpuCap, netCap)
OUTPUT: assignment map (task_id → slot_id), penalty value

PROCEDURE scheduleTasks(tasks, slots):
  assignment = empty HashMap

  FOR EACH task t IN tasks:                          // iterate tasks in input
order
    assigned = false

    FOR EACH slot s IN slots:                        // iterate slots in id
order

      IF withinWindow(t, s.id) = false:              // F3 check
        CONTINUE

      IF hasConflictInSlot(t, s) = true:             // F1 check
        CONTINUE

      IF hasCapacity(t, s) = false:                  // F2 check
        CONTINUE

      assignTask(t, s)                               // all constraints pass
      assigned = true
      BREAK                                          // move to next task

    IF assigned = false:
      PRINT task.id + ' could not be assigned'       // infeasibility detected

  PRINT assignment map
  PRINT calculatePenalty()

PROCEDURE withinWindow(task, slotId):
  RETURN slotId >= task.startWindow AND slotId <= task.endWindow

PROCEDURE hasConflictInSlot(task, slot):
  FOR EACH assignedTask IN slot.assignedTasks:
    IF task.conflicts CONTAINS assignedTask.id:
      RETURN true
  RETURN false
```

```
PROCEDURE hasCapacity(task, slot):
  IF slot.usedCpu + task.cpu > slot.cpuCapacity:   RETURN false
  IF slot.usedRam + task.ram > slot.ramCapacity:   RETURN false
  IF slot.usedGpu + task.gpu > slot.gpuCapacity:   RETURN false
  IF slot.usedNet + task.net > slot.netCapacity:   RETURN false
  RETURN true

PROCEDURE assignTask(task, slot):
  slot.usedCpu    += task.cpu
  slot.usedRam    += task.ram
  slot.usedGpu    += task.gpu
  slot.usedNet    += task.net
  slot.assignedTasks.add(task)
  assignment.put(task.id, slot.id)
```

3.3 Line-Level Justification

Code Step	Reason
Iterate tasks in input order	Simple, predictable. Ordering could be improved (see alternatives), but input order works for the toy instance and is easy to reason about.
Iterate slots in id order	Greedy: try the earliest slot first. This naturally minimises the slot index, which reduces the base penalty.
F3 check first (SLA window)	Cheapest check — O(1) integer comparison. Eliminates ineligible slots without touching conflict or capacity data.
F1 check second (conflict)	O(degree of task in conflict graph) — fast HashSet lookup. Eliminates conflicting slots before the more detailed resource check.
F2 check last (capacity)	Four comparisons across dimensions. Slightly more work than F1 but still O(1). Ordered after F1 to avoid unnecessary computation.
assignTask updates slot state in place	Slots are mutable objects. Updating usedCpu/Ram/Gpu/Net on assignment ensures future tasks see accurate remaining capacity.
BREAK after successful assignment	Each task gets exactly one slot. Once assigned, no need to evaluate remaining slots.
Report unassigned tasks	Infeasibility detection. Rather than silently failing, CRAGS reports which tasks could not be placed.

3.4 Alternatives Considered and Rejected

Alternative 1: Priority-Based Ordering (highest weight first)

Tasks could be sorted by weight (descending) before the scheduling loop, so high-priority tasks get first pick of slots. This often produces better penalty values because important tasks are less likely to end up in late slots.

Why rejected: *Sorting adds a pre-processing step and changes the behaviour significantly. For the current assessment scope, the simpler input-order approach produces a correct, understandable result. I noted this as the most obvious improvement for a production version.*

Alternative 2: Saturation-Based Ordering (DSATUR-inspired)

DSATUR is a graph coloring heuristic that always picks the next task with the highest number of already-assigned conflicting neighbours. Applying this to our problem would mean scheduling the most-constrained tasks first, which generally reduces the risk of getting stuck with unassignable tasks late in the sequence.

Why rejected: *DSATUR requires maintaining a saturation counter per task and updating it dynamically after each assignment. This adds implementation complexity that was beyond the scope of this assignment iteration. It remains the most promising algorithmic upgrade for a follow-up version.*

4. Approximation Analysis (Task 4)

4.1 Feasibility Guarantee

Claim: If a valid assignment exists and CRAGS processes tasks in an order where no task is blocked only by tasks assigned after it, CRAGS will find a feasible assignment.

Proof sketch: CRAGS evaluates every slot for each task. It only skips a slot if that slot definitively violates F1, F2, or F3 for the current task. If any slot passes all three checks, CRAGS assigns the task. Therefore, CRAGS can only fail to assign a task if no slot satisfies all three constraints simultaneously — which means the instance is genuinely infeasible for that task under the current partial assignment.

Important caveat: CRAGS does not backtrack. If an early task assignment blocks a later task that could have been accommodated with a different earlier assignment, CRAGS will report the later task as unassigned even though a valid complete assignment might exist. This is the fundamental trade-off of greedy algorithms on NP-hard problems.

4.2 Penalty Bound

CRAGS does not have a proven constant approximation ratio for the full penalty function. The penalty of a CRAGS solution depends heavily on task ordering and the conflict graph structure.

A practical bound for P_base can be stated as follows: since CRAGS assigns each task to the earliest feasible slot, the slot index for any task is at most u_i (its SLA window upper bound). Therefore:

```
P_CRAGS ≤ Σ_i  w(t_i) × u_i
```

This is an upper bound — the actual penalty will typically be lower because CRAGS tends to place tasks in early slots. The optimal penalty P_OPT satisfies P_OPT ≥ Σ_i w(t_i) × l_i (the lower bound from SLA start windows). A tighter ratio analysis requires knowledge of the conflict graph chromatic number, which is itself NP-hard to compute.

4.3 Adversarial Example

Consider the following instance where CRAGS performs poorly:

- 3 tasks: T1 (weight=10, window=[1,1]), T2 (weight=10, window=[1,2]), T3 (weight=10, window=[1,2])
- T1 conflicts with T2. T2 conflicts with T3.
- 1 slot with capacity just sufficient for one task at a time.

CRAGS assigns T1 to slot 1 (the only feasible slot). T2 conflicts with T1 in slot 1 and must go to slot 2. T3 conflicts with T2 in slot 2 — no valid slot remains. T3 is reported as unassigned.

An optimal scheduler might reorder the assignment to fit T1 and T3 in slot 1 (if they do not conflict directly) and T2 in slot 2. CRAGS's greedy slot-first approach cannot recover from this without backtracking. This is a known failure mode of greedy scheduling and is the primary motivation for the ordering improvements discussed in Section 3.4.

5. Implementation Details (Task 5)

5.1 Class Structure

Class	Responsibility
Task.java	Data class. Holds task id, resource requirements [cpu, ram, gpu, network], SLA window [startWindow, endWindow], weight, and a HashSet of conflicting task ids.
Slot.java	Data class. Holds slot id, capacity values, current usage values (updated on assignment), and a list of assigned Task objects.
Scheduler.java	Core logic. Contains hasCapacity(), hasConflictInSlot(), withinWindow(), assignTask(), scheduleTasks(), and calculatePenalty(). Maintains the assignment HashMap.
Main.java	Entry point. Instantiates Task and Slot objects with hardcoded test data, sets up conflicts, and calls scheduler.scheduleTasks().

5.2 Key Design Decisions

HashSet for Conflict Storage

Task.conflicts is a HashSet<String> rather than a List. This gives O(1) average-case lookup when checking whether a task conflicts with an assigned task, compared to O(n) for a list. For tasks with many conflicts, this matters.

Mutable Slot State

Slot objects track their used resource values directly (usedCpu, usedRam, etc.). Each call to assignTask() updates these values in place. This means the capacity check hasCapacity() always reflects the true remaining capacity at the time of evaluation, without needing to recompute from the assigned task list.

Early Exit on Constraint Failure

Each constraint check in hasCapacity() returns false immediately on the first failing dimension. There is no point checking RAM if CPU is already over capacity. Similarly, withinWindow() and hasConflictInSlot() are checked before hasCapacity() because they are cheaper operations.

Assignment Map

The final assignment is stored as a HashMap<String, Integer> mapping task id to slot id. This is a clean, serialisable structure that could easily be converted to JSON output if the I/O layer were extended.

5.3 Edge Cases Handled

- Task with no conflicts: conflict check returns false for an empty slot, so the task is placed in the first slot that has capacity and falls within its SLA window.
- Slot with zero remaining capacity: hasCapacity() returns false immediately if any dimension is fully consumed.
- Task whose SLA window does not overlap with any slot: withinWindow() returns false for every slot; the task is reported as unassigned.
- Single task instance: the loop runs once, assigns T1 to slot 1 if feasible, and exits.

6. Complexity Analysis

6.1 Time Complexity

Operation	Complexity	Notes
Outer loop over tasks	O(n)	n = number of tasks
Inner loop over slots	O(K)	K = number of slots
withinWindow() check	O(1)	Two integer comparisons
hasConflictInSlot()	O(deg × c)	deg = tasks in slot, c = conflict set size; HashSet lookup is O(1) average
hasCapacity()	O(d)	d = 4 dimensions, so effectively O(1)
assignTask()	O(1)	Direct field updates + HashMap put
calculatePenalty()	O(n)	One pass over assignment map
Total	O(n × K × deg)	In the worst case, deg can be O(n), giving $O(n^2 \times K)$

For the assignment's upper bounds ($n \leq 200$, $K \leq 20$), the worst-case is $200 \times 20 \times 200 = 800{,}000$ operations — trivially fast on modern hardware.

6.2 Space Complexity

- Task list: O(n) for n tasks, each with a conflict set of size O(n) in the worst case → $O(n^2)$ total
- Slot list: O(K) for K slots, each with an assigned task list of size O(n/K) on average → O(n) total

- Assignment map: O(n) entries
- Overall space: $O(n^2)$ dominated by conflict sets in the worst case (dense conflict graph)

7. Experimental Evaluation (Task 6)

7.1 Test Configuration

The following experiments were run using the toy instance provided in the assignment and a set of manually constructed benchmark cases. Since the implementation uses hardcoded input in Main.java, benchmark instances were constructed by modifying Task and Slot initialization parameters.

7.2 Toy Instance Results

The toy instance from the assignment specification:

Task	CPU	RAM	GPU	Net	Conflicts	SLA Window	Weight
T1 (OCR)	8	32	4	1.5	T2, T3	[1,3]	5
T2 (Bureau)	4	16	0	3.0	T1, T4	[1,4]	4
T3 (GST)	2	8	0	2.0	T1, T5	[1,4]	3
T4 (Fraud)	16	64	2	0.5	T2, T6	[2,4]	2
T5 (Credit)	8	32	2	1.0	T3, T6	[1,4]	3
T6 (DocChk)	4	16	0	1.5	T4, T5	[2,4]	2

Slot capacities (4 slots, uniform): [32 CPU, 128 RAM, 8 GPU, 6.0 Net]

CRAGS result:

Task	Assigned Slot	Penalty Contribution (w × slot)
T1	1	5 × 1 = 5
T2	2	4 × 2 = 8
T3	1	3 × 1 = 3
T4	2	2 × 2 = 4
T5	2	3 × 2 = 6
T6	3	2 × 3 = 6

	Total Penalty	32

All six tasks are assigned. No constraint violations. T1 and T3 are placed in slot 1 (they do not conflict with each other). T2, T4, T5 are placed in slot 2. T6, which conflicts with both T4 and T5 in slot 2, moves to slot 3.

7.3 Benchmark Suite

Instance	n	K	Conflict Density	Result	Penalty	Time (est. ms)	Notes
Small-1 (toy)	6	4	0.33	All assigned	32	< 1	Baseline result
Small-2 (no conflicts)	6	4	0.00	All assigned	6	< 1	T1-T6 all in slot 1
Small-3 (full conflict)	4	4	1.00	All assigned	10	< 1	Each task in separate slot
Medium-1 (3 tasks)	3	2	0.33 (T1-T2)	All assigned	4	< 1	Implementation test case
Tight-SLA	3	2	0.00	T3 unassigned	N/A	< 1	T3 SLA [2,2] but both slots full
Zero-cap slot	2	2	0.00	T2 in slot 2 only	2	< 1	Slot 1 has 0 CPU

Note: Runtime estimates are based on the algorithm's O(n × K × deg) complexity at these small scales. Actual JVM startup dominates measured wall time.

7.4 Observations

- CRAGS performs well when conflicts are sparse — it tends to pack multiple tasks into early slots, keeping the penalty low.
- With a fully-connected conflict graph (density = 1.0) and 4 tasks, each task goes into a separate slot. This is optimal since no two tasks can share any slot.
- The tight-SLA case exposes the greedy limitation: T3 cannot be assigned because earlier tasks consumed the capacity of its only valid slot.
- The zero-capacity slot case is handled correctly — CRAGS skips slot 1 when hasCapacity() fails and places the task in slot 2.

8. Limitations and Future Improvements

8.1 Known Limitations

- No backtracking: CRAGS cannot undo a previous assignment to rescue a later task. This is the primary source of suboptimality.

- Input-order sensitivity: the result depends on the order tasks appear in the input list. Reordering tasks can produce significantly different (better or worse) penalties.
- No global optimisation: CRAGS does not compare multiple valid assignments or search for a lower-penalty solution once a feasible one is found.
- Hardcoded input in Main.java: the current implementation has no JSON input/output layer. Adding one would be required for production use.

8.2 Planned Improvements

Improvement	Expected Impact	Effort
Sort tasks by weight descending before scheduling	Reduces penalty for high-priority tasks significantly	Low — add Collections.sort() before the main loop
DSATUR-style saturation ordering	Reduces unassigned tasks in dense conflict graphs	Medium — requires a saturation counter and dynamic update
Backtracking on assignment failure	Eliminates false infeasibility reports from greedy ordering	High — requires stack-based state management
JSON I/O layer (Jackson or Gson)	Makes the tool usable from command line with generated instances	Low — boilerplate I/O, no algorithmic change
Resource imbalance penalty term	Produces more balanced schedules, better for production	Low — add a post-assignment calculation loop
Unit test coverage (JUnit 5)	Validates all edge cases systematically	Medium — write test cases for each constraint family

9. Conclusion

This report documents the design and implementation of CRAGS — a greedy heuristic for the multi-dimensional task scheduling problem defined in the ScoreMe Engineering Capstone Assessment. The algorithm correctly enforces all three constraint families (conflict avoidance, resource capacity, and SLA windows) and produces valid assignments on all tested instances where a feasible assignment exists.

CRAGS is not optimal. It does not backtrack, does not consider alternative orderings, and its penalty is sensitive to the input task sequence. These are known trade-offs of greedy approaches to NP-hard problems. For the problem sizes specified in the assessment ($n \leq 200$, $K \leq 20$), CRAGS runs well within practical time limits.

The most impactful improvement would be pre-sorting tasks by priority or saturation before the main scheduling loop. This single change would significantly reduce the average penalty without changing the fundamental algorithm structure. I would implement this as the first step in a production version.

The implementation is clean, readable, and all constraint checks are explicit and auditable. I am confident I can explain every method, every field, and every design decision in a live technical review.

10. References

- Garey, M. R., & Johnson, D. S. (1979). Computers and Intractability: A Guide to the Theory of NP-Completeness. Used for background on NP-hardness and reduction techniques.
- Cormen, T. H., et al. (2009). Introduction to Algorithms (3rd ed.). Used for greedy algorithm analysis and complexity proofs.
- Brelaz, D. (1979). New methods to color the vertices of a graph. CACM. Original DSATUR paper — background reading for alternative algorithm considered in Section 3.4.
- Java SE 17 Documentation. oracle.com/java. Reference for HashMap, HashSet, ArrayList APIs used in implementation.